

Week – 5 Competitive Programming

Q1. Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5	1
1 1 2 3 4	

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2
3 int main(){
4     int n;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++){
8         scanf("%d",&arr[i]);
9     }
10    for(int i=0;i<n-1;i++){
11        for(int j=i+1;j<n;j++){
12            if(arr[i]==arr[j]){
13                printf("%d",arr[i]);
14                return 0;
15            }
16        }
17    }
18    return 0;
19 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q2. Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, a[1000], i, j;
5     scanf("%d", &n);
6
7     for (i = 0; i < n; i++) {
8         scanf("%d", &a[i]);
9     }
10
11     for (i = 0; i < n; i++) {
12         for (j = i + 1; j < n; j++) {
13             if (a[i] == a[j]) {
14                 printf("%d\n", a[i]);
15                 return 0;
16             }
17         }
18     }
19
20     return 0;
21 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q3. Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

• The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int t;
5     scanf("%d", &t);
6
7     while (t--) {
8         int n1, n2;
9         scanf("%d", &n1);
10
11         int a[n1];
12
13         for (int i = 0; i < n1; i++) {
14             scanf("%d", &a[i]);
15         }
16
17         scanf("%d", &n2);
18
19         int b[n2];
20
21         for (int i = 0; i < n2; i++) {
22             scanf("%d", &b[i]);
23         }
24
25         int i = 0, j = 0, first = 1;
26
27         while (i < n1 && j < n2) {
28             if (a[i] == b[j]) {
29                 if (!first) {
30                     printf(" ");
31                 }
32                 printf("%d", a[i]);
33                 first = 0;
34                 i++;
35                 j++;
36             } else if (a[i] < b[j]) {
37                 i++;
38             } else {
39                 j++;
40             }
41         }
42         printf("\n");
43     }
44
45     return 0;
46 }
47
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q4. Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:
 1. Line 1 contains N1, followed by N1 integers of the first array
 2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int t;
5     scanf("%d", &t);
6
7     while (t-->0) {
8         int n1, n2;
9         scanf("%d", &n1);
10
11         int a[n1];
12
13         for (int i = 0; i < n1; i++) {
14             scanf("%d", &a[i]);
15         }
16
17         scanf("%d", &n2);
18
19         int b[n2];
20
21         for (int i = 0; i < n2; i++) {
22             scanf("%d", &b[i]);
23         }
24
25         int i = 0, j = 0, first = 1;
26
27         while (i < n1 && j < n2) {
28             if (a[i] == b[j]) {
29                 if (!first)
30                     printf(" ");
31
32                 printf("%d", a[i]);
33                 first = 0;
34                 i++;
35                 j++;
36             } else if (a[i] < b[j]) {
37                 i++;
38             } else {
39                 j++;
40             }
41         }
42         printf("\n");
43     }
44
45     return 0;
46 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q5. Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, k;
5     scanf("%d", &n);
6
7     int a[n];
8
9     for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         int diff = a[j] - a[i];
19
20         if (i == j) {
21             j++;
22         } else if (diff == k) {
23             printf("1");
24             return 0;
25         } else if (diff < k) {
26             j++;
27         } else {
28             i++;
29         }
30     }
31
32     printf("0");
33
34     return 0;
35 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Q6. Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int n, k;
5     scanf("%d", &n);
6
7     int a[n];
8
9     for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         if (i == j) {
19             j++;
20         } else if (a[j] - a[i] == k) {
21             printf("1\n");
22             return 0;
23         } else if (a[j] - a[i] < k) {
24             j++;
25         } else {
26             i++;
27         }
28     }
29
30     printf("0\n");
31
32     return 0;
33 }

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.